Constructions With Monoidal Categories

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01UF This chapter contains some material on constructions with monoidal categories.

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01UG 12.1 Moduli Categories of Monoidal Structures

01UH 12.1.1 The Moduli Category of Monoidal Structures on a Category

Let $\mathcal{C}$ be a category.

01UJ **Definition 12.1.1.1.1.** The **moduli category of monoidal structures on $\mathcal{C}$** is the category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ defined by

$$\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \overset{\text{def}}{=} \mathsf{pt} \times_{\mathsf{Cats}} \mathsf{MonCats}, \qquad \begin{tikzcd} \mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathsf{MonCats} \arrow[d, "\mathsf{up}"] \\ \mathsf{pt} \arrow[r, "{[\mathcal{C}]}"'] & \mathsf{Cats}. \end{tikzcd}$$

01UK **Remark 12.1.1.1.2.** In detail, **the moduli category of monoidal structures on $\mathcal{C}$** is the category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ where:

- *Objects.* The objects of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ are monoidal categories $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ whose underlying category is $\mathcal{C}$.

- *Morphisms.* A morphism from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ is a strong monoidal functor structure

$$\begin{aligned} \mathrm{id}^{\otimes}_{\mathcal{C}} &\colon A \boxtimes_{\mathcal{C}} B \overset{\sim}{\dashrightarrow} A \otimes_{\mathcal{C}} B, \\ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}} &\colon \mathbb{1}'_{\mathcal{C}} \overset{\sim}{\dashrightarrow} \mathbb{1}_{\mathcal{C}} \end{aligned}$$

  on the identity functor $\mathrm{id}_{\mathcal{C}} \colon \mathcal{C} \to \mathcal{C}$ of $\mathcal{C}$.

- *Identities.* For each $M \overset{\text{def}}{=} \left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right) \in \mathrm{Obj}(\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}))$, the unit map

$$\mathbb{1}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,M} \colon \mathrm{pt} \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, M)$$

  of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ at M is defined by

$$\mathrm{id}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M} \overset{\text{def}}{=} \left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right),$$

  where $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ is the identity monoidal functor of $\mathcal{C}$ of **??**.

- *Composition.* For each $M, N, P \in \mathrm{Obj}(\boldsymbol{\mathcal{M}}_{\mathbb{E}_1}(\boldsymbol{\mathcal{C}}))$, the composition map

$$\circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} \colon \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(N, P) \times \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, N) \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, P)$$

of $\boldsymbol{\mathcal{M}}_{\mathbb{E}_1}(\boldsymbol{\mathcal{C}})$ at (M, N, P) is defined by

$$\left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}}\right) \circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} \left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right) \overset{\mathrm{def}}{=} \left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right).$$

01UL **Remark 12.1.1.1.3.** In particular, a morphism in $\boldsymbol{\mathcal{M}}_{\mathbb{E}_1}(\boldsymbol{\mathcal{C}})$ from $\left(\boldsymbol{\mathcal{C}}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\boldsymbol{\mathcal{C}}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ satisfies the following conditions:

01UM 1. *Naturality.* For each pair $f \colon A \to X$ and $g \colon B \to Y$ of morphisms of $\boldsymbol{\mathcal{C}}$, the diagram

$$\begin{tikzcd}
A \boxtimes_{\mathcal{C}} B \arrow[r, "f \boxtimes_{\mathcal{C}} g"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B}"'] & X \boxtimes_{\mathcal{C}} Y \arrow[d, "\mathrm{id}^{\otimes}_{X,Y}"] \\
A \otimes_{\mathcal{C}} B \arrow[r, "f \otimes_{\mathcal{C}} g"'] & X \otimes_{\mathcal{C}} Y
\end{tikzcd}$$

commutes.

01UN 2. *Monoidality.* For each $A, B, C \in \mathrm{Obj}(\boldsymbol{\mathcal{C}})$, the diagram

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"'] & & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

commutes.

01UP 3. *Left Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} A \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}'_{\mathcal{C}},A}"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} A \arrow[dr, "\lambda^{\mathcal{C}}_{A}"] & \\
\mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} A \arrow[ur, "\mathrm{id}^{\otimes}_{\mathbb{1}} \boxtimes_{\mathcal{C}} \mathrm{id}_A"] \arrow[rrr, "\lambda^{\mathcal{C},\prime}_{A}"'] & & & A
\end{tikzcd}$$

commutes.

01UQ 4. *Right Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & A \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{A,\mathbb{1}'_{\mathcal{C}}}"] & A \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[dr, "\rho^{\mathcal{C}}_{A}"] & \\
A \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[ur, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[rrr, "\rho^{\mathcal{C},\prime}_{A}"'] & & & A
\end{tikzcd}$$

commutes.

01UR **Proposition 12.1.1.1.4.** Let $\mathcal{C}$ be a category.

01US 1. *Extra Monoidality Conditions.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

01UT (a) The diagram

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"'] & (A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & (A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C
\end{tikzcd}$$

commutes.

01UU (b) The diagram

$$\begin{tikzcd}
A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

commutes.

01WB 2. *Extra Monoidal Unity Constraints.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

01WC (a) The diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes.

01WD (b) The diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes.

01WE (c) The diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes.

01WF (d) The diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes.

01UV 3. *Mixed Associators.* Let $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ and $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ be monoidal structures on $\mathcal{C}$ and let

$$\mathrm{id}^{\otimes}_{-_1,-_2} \colon -_1 \boxtimes_{\mathcal{C}} -_2 \to -_1 \otimes_{\mathcal{C}} -_2$$

be a natural transformation.

01UW (a) If there exists a natural transformation

$$\alpha^{\otimes}_{A,B,C} \colon (A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \to A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_{A,B \boxtimes_{\mathcal{C}} C}"] \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\otimes}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

01UX (b) If there exists a natural transformation

$$\alpha^{\boxtimes}_{A,B,C} \colon (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \to A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A,B \otimes_{\mathcal{C}} C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B,C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

01UY (c) If there exists a natural transformation

$$\alpha^{\boxtimes,\otimes}_{A,B,C} \colon (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \to A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B,C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A,B \boxtimes_{\mathcal{C}} C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

Proof. *Item 1, Extra Monoidality Conditions*: We claim that Items 1a and 1b are indeed true:

1. *Proof of Item 1a:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms $\mathrm{id}^{\otimes}_{A,B}$ and id_C.

2. *Proof of Item 1b:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms id_A and $\mathrm{id}^{\otimes}_{B,C}$.

This finishes the proof.
Item 2, Extra Monoidal Unity Constraints: We claim that Items 2a and 2b are indeed true:

1. *Proof of Item 1a:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[rr, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[dr, "\mathrm{id}_{\mathbb{1}_{\mathcal{C}}} \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[dd, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\mathrm{id}_{\mathbb{1}_{\mathcal{C}}} \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[ddd, bend left, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dd, equal] \arrow[dr, phantom, "(1)"] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(2)"] & \\
\mathbb{1}'_{\mathcal{C}} \arrow[drr, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] \arrow[dr, phantom, "(3)"] & & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(4)"] & \\
& & \mathbb{1}_{\mathcal{C}} & &
\end{tikzcd}$$

whose boundary diagram is the diagram whose commutativity we wish to prove. Since:

- Subdiagram (1) commutes by the naturality of $\mathrm{id}^{\otimes,-1}_{\mathcal{C}}$;
- Subdiagram (2) commutes trivially;

- Subdiagram (3) commutes by the naturality of $\lambda^{\mathcal{C}}$, where the equality $\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}$ comes from **??**;
- Subdiagram (4) commutes by the right monoidal unity of $\left(\mathrm{id}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathcal{C}|\mathbb{1}}\right)$;

so does the boundary diagram, and we are done.

2. *Proof of Item 1b:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[rr, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}'_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, "\mathrm{id}^{\otimes}_{\mathbb{1}} \otimes_{\mathcal{C}} \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dd, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[ddrr, bend right, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] & & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}} \boxtimes_{\mathcal{C}} \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[ddd, "\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dd, equal] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[drr, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d] \\
& & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

(1) (2) (3) (4)

whose boundary diagram is the diagram whose commutativity we wish to prove. Since:

- Subdiagram (1) commutes by the naturality of $\mathrm{id}_{\mathcal{C}}^{\otimes,-1}$;
- Subdiagram (2) commutes trivially;
- Subdiagram (3) commutes by the naturality of $\rho^{\mathcal{C}}$, where the equality $\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}$ comes from **??**;
- Subdiagram (4) commutes by the left monoidal unity of $\left(\mathrm{id}_{\mathcal{C}}, \mathrm{id}_{\mathcal{C}}^{\otimes}, \mathrm{id}_{\mathcal{C}|\mathbb{1}}^{\otimes}\right)$;

so does the boundary diagram, and we are done.

3. *Proof of Item 2c:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[dr, phantom, "(1)"] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}.
\end{tikzcd}$$

Since:

- The boundary diagram commutes trivially;
- Subdiagram (1) commutes by Item 1b;

it follows that the diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes. But since $\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}$ is an isomorphism, it follows that the diagram (†) also commutes, and we are done.

4. *Proof of Item 2d:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[dr, phantom, "(1)"] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

Since:

- The boundary diagram commutes trivially;
- Subdiagram (1) commutes by Item 1a;

it follows that the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes. But since $\mathrm{id}^{\otimes,-1}_{\mathbb{1}}$ is an isomorphism, it follows that the diagram (†) also commutes, and we are done.

This finishes the proof.

Item 3, Mixed Associators: We claim that Items 3a to 3c are indeed true:

01UZ 1. *Proof of Item 3a:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$

of Item 2 of Definition 12.1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] \arrow[dd, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & (1) & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dd, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
 & (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dl, dashed, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, dashed, "\alpha^{\otimes}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] & (3) & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by Item 1a of Item 1.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by assumption.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

01V0 2. *Proof of Item 3b:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of Item 2 of Definition 12.1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dr, dashed, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & (1) & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dl, dashed, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"'] \arrow[dd, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
 & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dd, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] & (2) \quad (3) & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by assumption.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by Item 1b of Item 1.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

01V1 3. *Proof of Item 3c:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of Item 2 of Definition 12.1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] \arrow[dr, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[r, phantom, "(1)"] & & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] \arrow[r, phantom, "(2)"] & & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since subdiagrams (1) and (2) commute by assumption, it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 12.1.1.1.3.

This finishes the proof. □

01V2 ### 12.1.2 The Moduli Category of Braided Monoidal Structures on a Category

01V3 ### 12.1.3 The Moduli Category of Symmetric Monoidal Structures on a Category

01V4 ## 12.2 Moduli Categories of Closed Monoidal Structures

01V5 ## 12.3 Moduli Categories of Refinements of Monoidal Structures

01V6 ### 12.3.1 The Moduli Category of Braided Refinements of a Monoidal Structure

Appendices

A Other Chapters

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2. A Guide to the Literature

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3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

Relations

8. Relations
9. Constructions With Relations
10. Conditions on Relations

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11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes